Vector 3D

Mind map

Vector basics

- $\hat{a} = \vec{a}/|\vec{a}|$
- Collinear: $\vec{a} = \lambda\vec{b}$
- Angle $0 \le \theta \le \pi$, tails together

Addition

- Closed figure sum $= \vec{0}$
- Diagonals $\vec{a} + \vec{b}$, $\vec{b} - \vec{a}$
- $|\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$

3D coordinates

- Section $\dfrac{m\vec{b} + n\vec{a}}{m+n}$
- From x-axis $\sqrt{y^2 + z^2}$

Linear dependence

- Plane: $\vec{r} = x\vec{a} + y\vec{b}$
- Independent: compare coefficients
- Collinear points: $\sum x = 0$

Dot product

- $|\vec{a}||\vec{b}|\cos\theta$; $\perp \Leftrightarrow 0$
- Projection $\vec{a} \cdot \hat{b}$
- $|\vec{a} \pm \vec{b}|^2$ expansion

Cross product

- Determinant; $\parallel \Leftrightarrow \vec{0}$
- Area $\frac{1}{2}|\vec{AB} \times \vec{AC}|$
- Lagrange identity

DC and DR

- $l^2 + m^2 + n^2 = 1$
- Divide DR by $\sqrt{a^2 + b^2 + c^2}$
- $\cos\theta = |\vec{b}_1 \cdot \vec{b}_2|/(|\vec{b}_1||\vec{b}_2|)$

Line

- $\vec{r} = \vec{a} + t\vec{b}$; general point
- Foot: $\vec{PQ} \cdot \vec{b} = 0$
- Bisector $\hat{a} + \hat{b}$

Two lines

- Intersect: STP $= 0$
- SD $\dfrac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$
- Parallel: $\dfrac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

Plane forms

- $\vec{r} \cdot \hat{n} = d$; $ax + by + cz + d = 0$
- Three points: determinant
- Intercept $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Angles

- Plane-plane: cos of normals
- Line-plane: $\sin\theta$
- Line in plane: $\vec{b} \cdot \vec{n} = 0$ + point

Distance and image

- $\dfrac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$
- Foot $k = 1$, image $k = 2$
- Bisector planes: $d_1, d_2 > 0$

Family and intersection

- $P_1 + \lambda P_2 = 0$
- Direction $\vec{n}_1 \times \vec{n}_2$
- Image of line: two points
- Greatest slope $(\vec{n}_H \times \vec{n}_G) \times \vec{n}_G$

Scalar triple product

- Determinant, cyclic
- Volume $|[\vec{a}\ \vec{b}\ \vec{c}]|$, $\frac{1}{6}$ tetrahedron
- $= 0 \Leftrightarrow$ coplanar

Vector triple product

- $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
- Four vectors: dot / cross of crosses
- In plane of $\vec{p}, \vec{q}$, $\perp \vec{r}$: $\vec{r} \times (\vec{p} \times \vec{q})$

Space combination, equations

- $x = [\vec{r}\ \vec{b}\ \vec{c}]/[\vec{a}\ \vec{b}\ \vec{c}]$
- Reciprocal $\vec{a}' = \vec{b} \times \vec{c}/[\vec{a}\ \vec{b}\ \vec{c}]$
- Dot or cross with a known vector